

CAIRNWELL, United Kingdom

WMO: 030720

Lat: 56.8795N

Lon: 3.4213W

Elev: 3061

StdP: 13.14

Time Zone: 0.00 (EUW)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.7	19.4	7.2	8.9	33.9	13.5	12.2	26.6	71.5	31.6	65.4	30.1	30.7	350	1.208

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.4	61.7	53.8	58.7	52.5	56.3	51.6	56.6	58.8	54.7	56.7	53.0	54.9	14.2	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
55.6	73.8	57.4	53.8	68.9	55.3	52.0	64.6	53.7	25.5	59.0	24.1	56.6	23.1	55.2	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 81.5	64.3	54.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			DB	15.0	58.5	2.3	2.0	13.4	59.9	12.0	61.1	10.7	62.2	9.0	63.7	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.8	29.4	29.7	30.9	34.9	40.8	44.6	48.6	47.3	44.5	39.1	32.8	30.8		
	DBStd	8.76	4.88	5.78	5.88	6.10	6.31	5.44	5.29	4.42	4.57	4.93	5.00	5.41		
	HDD50	4556	640	568	593	454	297	178	93	109	174	339	517	594		
	HDD65	9918	1105	988	1057	903	751	612	509	550	614	803	967	1059		
	CDD50	109	0	0	0	0	11	16	48	24	9	1	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	2	0	0	0	0	0	2	0	0	0	0	0	0		
	CDH80	1	0	0	0	0	0	1	0	0	0	0	0	0		
	WSAvg	24.8	28.0	28.4	27.2	23.5	19.2	17.2	17.0	21.1	24.3	28.4	29.6	33.5		
	PrecAvg	38.9	4.6	3.1	2.6	2.4	2.5	3.0	3.1	2.7	4.3	4.2	3.8			
(15) Precipitation	PrecMax	49.9	9.3	7.8	5.7	5.9	4.6	5.0	6.0	7.3	6.9	7.4	7.8	10.9		
	PrecMin	25.2	0.5	0.8	0.4	0.8	0.7	0.8	0.9	0.4	0.6	0.8	1.5	1.1		
	PrecStd	5.3	2.2	1.8	1.0	1.3	1.2	1.1	1.3	1.7	1.6	1.8	1.8	2.1		
	PrecStd	5.3	2.2	1.8	1.0	1.3	1.2	1.1	1.3	1.7	1.6	1.8	1.8	2.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.4	46.5	50.4	54.1	61.9	63.8	65.9	64.2	59.2	54.0	48.5	47.4		
		MCWB	38.5	38.9	39.4	44.4	52.1	51.8	56.7	56.5	49.3	43.5	45.3	42.1		
	2%	DB	41.2	43.2	44.4	49.6	57.1	58.1	62.2	59.4	55.8	50.1	44.9	44.4		
		MCWB	40.2	40.2	38.6	43.8	50.3	50.8	54.9	55.1	52.1	49.1	43.8	41.2		
	5%	DB	39.2	40.3	41.1	46.4	53.5	55.0	59.2	56.1	53.7	48.3	42.5	41.7		
		MCWB	38.7	38.9	38.2	41.5	47.1	50.1	53.3	53.0	51.1	47.2	41.0	40.2		
	10%	DB	37.0	38.0	38.8	43.4	50.4	52.2	56.7	53.5	51.2	46.2	40.4	38.8		
		MCWB	36.5	37.0	37.3	40.1	44.6	48.4	52.6	51.4	48.8	45.5	39.3	38.0		
	0.4%	WB	42.8	44.8	42.7	48.2	55.0	55.6	58.8	59.8	56.1	50.7	47.2	45.8		
		MCDB	42.9	45.0	45.7	49.9	58.8	59.0	62.2	61.9	57.2	50.8	47.6	46.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.4	41.9	40.4	45.2	51.8	53.2	56.7	56.6	53.8	49.0	43.7	43.1		
		MCDB	40.7	42.2	42.4	47.7	56.0	55.4	59.4	57.9	54.4	49.3	44.2	43.5		
	5%	WB	38.5	39.5	38.8	42.5	48.4	51.2	54.9	53.7	51.3	46.9	41.4	40.6		
		MCDB	38.9	39.9	40.4	45.1	52.1	53.6	57.4	55.0	52.3	47.3	42.1	41.2		
	10%	WB	36.5	37.2	37.3	40.6	45.9	49.2	53.1	52.0	49.1	45.1	39.4	37.9		
		MCDB	37.0	37.5	38.9	42.8	49.1	51.5	55.6	53.2	50.1	45.5	40.0	38.3		
	5% DB	MDBR	6.0	6.3	6.7	7.8	9.6	8.8	8.4	7.4	7.0	6.3	6.1	6.6		
		MCDBR	9.6	9.2	9.3	11.6	11.8	12.0	10.8	9.6	9.3	9.2	9.9	10.4		
		MCWBR	9.0	7.8	7.0	8.0	7.7	8.5	7.4	7.4	8.5	8.8	9.5	10.0		
		MCWBR	9.3	8.4	9.0	9.7	11.6	10.3	9.3	8.4	8.8	8.3	9.9	10.1		
(35) Mean Daily Temperature Range	5% WB	MCWBR	9.1	8.2	7.9	7.7	8.3	8.4	7.6	7.5	8.6	8.5	9.8	10.3		
		MCWBR	9.1	8.2	7.9	7.7	8.3	8.4	7.6	7.5	8.6	8.5	9.8	10.3		
	Clear-Sky Solar Irradiance	taub	0.267	0.288	0.311	0.330	0.331	0.330	0.331	0.320	0.308	0.296	0.269	0.249		
		taud	2.299	2.421	2.405	2.402	2.435	2.452	2.446	2.485	2.532	2.520	2.383	2.227		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	201	250	271	282	288	289	286	283	271	245	201	172		
		Edh at Noon	15	21	27	32	33	33	32	29	24	19	14	12		
(44) All-Sky Solar Radiation	RadStd	RadAvg	148	354	679	1112	1493	1471	1423	1126	790	421	193	106		
		RadStd	21	32	83	114	86	122	120	74	58	43	19	13		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page